

Answer all questions TIME: 1:30mins

INDEX NUMBER

QUESTION ONE

Electrolytes are regulated mainly by the kidneys and the endocrine system and the gastrointestinal system. Through the process of glomerular filtration, tubal reabsorption and excretion.

Human beings require some electrolyte in the blood to make life possible.

- (a) Copy and complete the table with the normal ranges of the following electrolyte listed below in mmol/L

ELECTROLYTE	NORMAL RANGES
1. Sodium	(135 - 145) mmol/L
2. Potassium	(3.5 to 5.5) mmol/L
3. Calcium	(2.0 to 2.6) mmol/L
4. Magnesium	(0.7 to 1.0) mmol/L
5. Phosphorus	(0.8 to 1.6) mmol/L
6. Chloride	(98 to 106) mmol/L

(3marks)

- (b) Metabolic acidosis is a clinical disturbance characterized by a low pH and low plasma bicarbonate

- (i) List Eight causes of metabolic acidosis (4marks)
 (ii) List Six clinical manifestations of metabolic acidosis (3mark)

- (c) Respiratory acidosis is a clinical condition in which the pH is less than 7.35 and PaCO₂ is greater than 42 mmHg

- (i) State Eight (8) causes of respiratory acidosis (4marks)
 (ii) List the treatment of respiratory acidosis (4marks)
 (iii) Define respiratory alkalosis (2marks)

renal failure

prolong use of diuretics

Aspirin poisoning

Diarrhoea

overdose of sedative

Excessive administration of chloride.

1. The normal duration for capillary refilling is second (s)
 - A. less than one
 - B. less than ten
 - C. less than three
 - D. more than ten
2. Which of the following techniques is used to assess air capacity of the lungs?
 - A. Auscultation
 - B. Inspection
 - C. Palpation
 - D. Percussion
3. When assessing the chest with a stethoscope, a lub-dub sound is heard. This sound is caused by closure of the:
 - A. atrioventricular valves and opening of the semilunar valves respectively
 - B. atrioventricular valves and semilunar valves respectively
 - C. atrioventricular valves and semilunar valves simultaneously
 - D. semilunar valves and atrioventricular valves respectively
4. The normal heart sound 'S2' occurs with the:
 - A. closure of atrioventricular valve and signals the end of systole
 - B. closure of semilunar valves and signals the end of systole
 - C. opening of semilunar valves and signals the beginning of systole
 - D. opening of the atrioventricular valve and signals opening of systole
5. During assessment of a patient, the following outcomes will be best assessed using palpation technique except:
 - A. moisture
 - B. sound
 - C. temperature
 - D. texture

The nurse is documenting the findings from a health assessment. Which of the following demonstrates the documentation of subjective information?

- A. Abdomen soft and non-tender to palpation
- B. Blood pressure 110/68mmHg
- C. "It hurts when I put weight on my leg".
- D. "Pulses present in lower extremities".

Which of the following is considered an example of objective data?

- A. A sore throat
- B. Alert and oriented
- C. An earache
- D. Dizziness

8. After completing the health history, the nurse begins to ask more detailed questions to clarify points and follow up on concerns expressed by the client during the interview. This portion of the health assessment is:

- A. focused interview
- B. informal teaching
- C. interpretation of findings
- D. objective data

9. A patient admitted to the hospital with asthma presented with the following problems based on an admission health history and physical assessment. Which problem is a first level priority?

- A. Impaired gas exchange
- B. Ineffective self-health management
- C. Readiness for enhanced spiritual well-being
- D. Risk for infection

10. You are taking a health history on a patient and noticed the eyes seem a bit yellow and the skin jaundiced. What word/phrase best describes this type of objective observation?

- A. A complication
- B. A sign
- C. A symptom
- D. An emergency

11. A rubbing, grating or squeaky sound heard on auscultation as if two pieces of leather are being rubbed together is described as:

- A. pleural friction rub
- B. pulmonary bruising
- C. pulmonary friction
- D. whooping cough

12. Which of these presentations on assessment sounds like snoring and it is caused by mucus accumulation in the trachea or large bronchi?

- A. Crackles
- B. Rhonchi
- C. Stridor
- D. Wheezes

13. This is an abnormal respiration cycle beginning with slow, shallow respiration that becomes rapid, then slower, followed by periods of apnea that last for about 20 seconds.
- Botte's respiration
 - Cheyne-stokes respiration
 - Kussmaul's respiration
 - Whooping respiration
14. When conducting a respiratory assessment on a patient, which of the following is the most appropriate technique?
- Allow a patient with a slight lower back pain to lie supine on bed
 - Instruct patient to breathe through the mouth quietly, more deeply and slowly
 - Listen to the heart sound at the same time that you are listening to the lung sounds
 - Use a stethoscope over the clothing of a patient who feels cold
15. Which type of the database is most appropriate for an individual who is admitted to a long-term care facility?
- Complete
 - Emergency
 - Episodic
 - Follow-up
16. Which of the following assessment uses the technique of percussion?
- Placing a stethoscope on the abdomen to assess the extent of bowel sounds
 - Pressing down on the symphysis pubis to determine the presence of urinary retention
 - Touching the skin over an artery to obtain a pulse rate
 - Using a rubber-tipped hammer to elicit a deep tendon reflex
17. While performing a cardiac assessment on a client with an incompetent heart valve, the nurse auscultates and heard a murmur. Which of the following best describes the sound of a heart murmur?
- Abrupt, high-pitched snapping noise.
 - Gentle, blowing or swooshing noises
 - Lub-dub sounds.
 - Scratchy, leathery heart noise
18. A plan of care that identifies the specific needs of the client as well as the needs which addressed by the skilled nursing facility is:
- health assessment
 - health examination
 - health identification
 - patient's assessment

25. When reviewing a medical record you see the patient has stated "I have had a cough for the last 10 days and now my chest feels very tight". This is an example of:

- A. chief complaint
- B. functional assessment
- C. past medical history
- D. recording of statement

26. Which of the following is an example of subjective data? The patient:

- A. has a BP of 110/40 mmHg
- B. has a pulse of 79bpm
- C. presents with shaky hands
- D. tells you she has severe migraine

27. During a health interview, the client states that she becomes increasingly short of breath when sitting in city traffic. The nurse views this information as a /anfactor.

- A. external environmental
- B. internal environmental
- C. psychoemotional
- D. sociocultural

28. During the process of physical assessment of a patient, which of the following would alert the nurse of a possible fluid volume deficit?

- A. Abdominal distention with tenderness
- B. Lips and mucous membranes dry
- C. Nausea, anorexia and agitation
- D. Skin warm, moist and elastic

29. In the process of interviewing during assessment, it was realised that patient does not seem to be giving reliable information, what will you do as a nurse?

- A. Call the family member listed on their intake chart.
- B. Continue to ask the same questions intermittently but rephrase them.
- C. Have the patient fill out their own printed health history
- D. Review their previous health records if available.

A medical diagnosis is used to evaluate a:

- A. cause of disease state in a person
- B. person's cultural practises and belief
- C. person's state of health
- D. response to actual or potential health problems

37. All of the following would occur if there was an increase in plasma osmolality:

- A. production of concentrated urine
- B. production of large volume of urine
- C. release of antidiuretic hormone
- D. stimulation of the thirst centre

38. Water gained by the body through metabolic activities is referred to as:

- A. insensible water gain
- B. metabolic water gain
- C. sensible water gain
- D. water gain by oxidation

39. When the concentration of Na^+ in the extracellular fluid decreases:

- A. a person experiences an increased thirst
- B. aldosterone level will increase
- C. more ADH is released
- D. osmoreceptors are stimulated

40. In a dehydrated state, antidiuretic hormone will act on the to water excretion.

- A. intestines; decrease
- B. intestines; increase
- C. kidneys; decrease
- D. kidneys; increase

41. A patient developed hypervolaemia as a result of fluid and sodium retention. These factors can cause water and sodium retention in the body except:

- A. blood replacement
- B. corticosteroid therapy
- C. hyperaldosteronism
- D. low intake of protein

42. In hypovolaemia, a 25% of intravascular volume loss will make the patient present the following clinical manifestations except:

- A. becoming more confused
- B. becoming more irritable
- C. skin may feel cool and clammy
- D. urine output is less than 10 ml/hour

- The body of a patient with hypovolaemia will activate compensatory mechanisms to maintain homeostasis by:
- causing orthostatic hypotension
 - increasing capillary refilling
 - increasing heart rate
 - increasing urine production
44. A patient with dehydration was admitted when he started exhibiting neurological presentations. The following are neurological presentations in dehydration except:
- excessive thirst
 - increase heart rate
 - loss of consciousness
 - skeletal muscle weakness
45. A patient on admission has been diagnosed with fluid volume deficit (hypovolaemia) as a result of third space shifting. The following conditions will result in third space shifting except:
- crush injuries
 - heart failure
 - hip fracture
 - vomiting
- Whenever the rate of sodium intake or output changes, there is a corresponding gain or loss of water that tends to:
- alter the sodium concentration
 - decrease the sodium concentration
 - increase the sodium concentration
 - keep the sodium concentration constant
- All of the homeostatic mechanisms that monitor and adjust the composition of body fluids respond to changes in the:
- extracellular fluid
 - fluid balance
 - intracellular fluid
 - regulatory hormones
- As a result of the aging process, the ability to regulate pH through renal compensation changes as a result of:
- reduction in the number of functional nephrons
 - reduction in the rate of insensible perspiration
 - increased ability to concentrate urine
 - increased glomerular filtration

49. The exchange between plasma and interstitial fluid is determined by the relationship between the:
- fluid balance and acid-base balance
 - net hydrostatic and net colloid osmotic pressures
 - osmotic pressure and cardiac output
 - total peripheral resistance and mean arterial pressure
50. The sodium-potassium pump transports sodium ions:
- into and out of cells in equal amounts
 - into cells
 - into skeletal muscles
 - out of cells
51. Which of the following can decrease the rate of calcium absorption in the human body?
- Acidic medium
 - Alkaline medium
 - High dietary fibre
 - Increase fluid concentration
52. Anane is diagnosed with fluid volume deficit. Which of the following nursing diagnosis might apply to his condition?
- Altered gastric activity
 - Altered urinary elimination
 - Decreased cardiac output
 - Increased cardiac output
53. A disease of which of the following structures is **most** likely to affect electrolyte reabsorption?
- Bladder
 - Glomerulus
 - Renal pelvis
 - Renal tubules
54. When teaching a patient about foods high in magnesium, the nurse would include:
- butter
 - cheese
 - green vegetables
 - tomatoes

55. When serum calcium levels rise, which of the following hormones is secreted to normalise the level in the blood?
- A. Aldosterone
 - B. Calcitonin
 - C. Parathyroid hormone
 - D. Renin
56. Which of the following diagnoses is most appropriate for a patient with hypocalcaemia?
- A. Airway clearance, ineffective
 - B. Constipation, bowel
 - C. High risk for injury: bleeding
 - D. High risk for injury: confusion
57. A patient with which of the following disorders is at high risk for developing hyperphosphataemia?
- A. Hyperglycaemia
 - B. Hyperkalaemia
 - C. Hypocalcaemia
 - D. Hyponatraemia
58. Which of the following statements provides the rationale for using a hypotonic solution for a patient with fluid volume deficit?
- A. A hypotonic solution is used to treat patients with hyponatraemia.
 - B. A hypotonic solution provides free water for the kidneys eliminate solutes.
 - C. A hypotonic solution supplies an excess of sodium and chloride ions.
 - D. Excessive volumes are recommended in the early postoperative period.
59. Mr. Adams is scheduled to receive an isotonic solution; which one of the following is an example of such a solution?
- A. 0.045% saline
 - ☒ B. 0.9% saline
 - C. 3% normal saline
 - D. D10% water
60. John is admitted to the hospital and is currently receiving hypertonic fluid. Which of the following management for the client includes monitoring for all of the following complications except:
- A. cellular shrinkage
 - B. cellular dehydration
 - C. fluid volume excess (FVE)
 - D. water intoxication